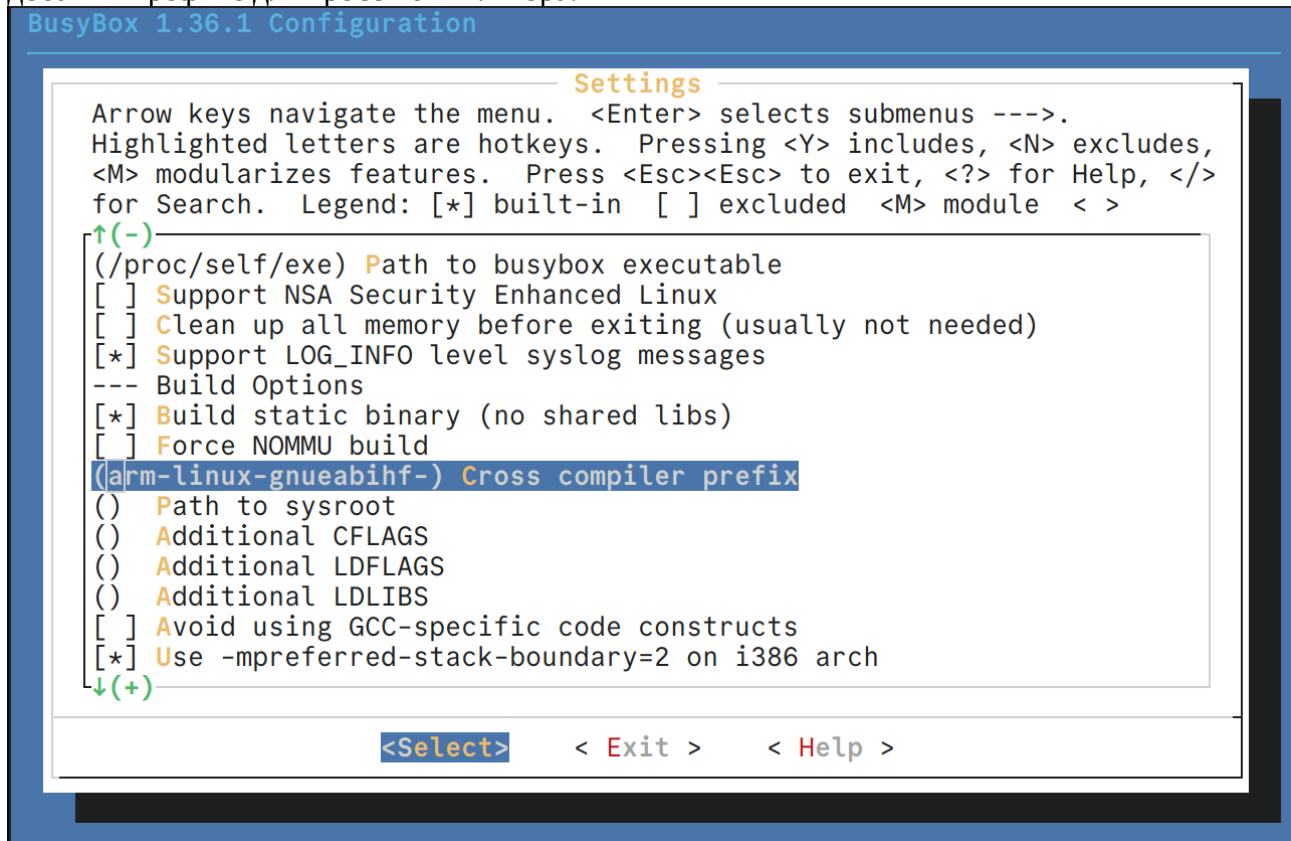


Создаём конфиг по умолчанию под ARM для Busybox:

```
vadim@vadim-PC:~/Repos/busybox-1.36.1$ make mrproper
vadim@vadim-PC:~/Repos/busybox-1.36.1$ ARCH=arm make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/basic/split-include
HOSTCC scripts/basic/docproc
GEN include/applets.h
GEN include/usage.h
GEN findutils/Kbuild
GEN findutils/Config.in
syslogd (13 kb) (SYSLOGD) [Y/n/?] (NEW) y
Rotate message files (FEATURE_ROTATE_LOGFILE) [Y/n/?] (NEW) y
Remote Log support (FEATURE_REMOTE_LOG) [Y/n/?] (NEW) y
Support -D (drop dups) option (FEATURE_SYSLOGD_DUP) [Y/n/?] (NEW) y
Support syslog.conf (FEATURE_SYSLOGD_CFG) [Y/n/?] (NEW) y
Include milliseconds in timestamps (FEATURE_SYSLOGD_PRECISE_TIMESTAMPS) [N/y/?] (NEW) n
Read buffer size in bytes (FEATURE_SYSLOGD_READ_BUFFER_SIZE) [256] (NEW) 256
Circular Buffer support (FEATURE_IPC_SYSLOG) [Y/n/?] (NEW) y
Circular buffer size in Kbytes (minimum 4KB) (FEATURE_IPC_SYSLOG_BUFFER_SIZE) [16] (NEW) 16
Linux kernel printk buffer support (FEATURE_KMSG_SYSLOG) [Y/n/?] (NEW) y
vadim@vadim-PC:~/Repos/busybox-1.36.1$
```

Через ARCH=arm make menuconfig укажем параметр для статической сборки и добавим префикс для кросс-компилятора:



Сохраняем конфигурацию и начинаем сборку:

```

vadim@vadim-PC:~/Repos/busybox-1.36.1$ ARCH=arm make menuconfig
HOSTLD  scripts/kconfig/mconf
HOSTCC  scripts/kconfig/lxdialog/checklist.o
HOSTCC  scripts/kconfig/lxdialog/inputbox.o
HOSTCC  scripts/kconfig/lxdialog/lxdialog.o
HOSTCC  scripts/kconfig/lxdialog/menubox.o
HOSTCC  scripts/kconfig/lxdialog/msgbox.o
HOSTCC  scripts/kconfig/lxdialog/textbox.o
HOSTCC  scripts/kconfig/lxdialog/util.o
HOSTCC  scripts/kconfig/lxdialog/yesno.o
HOSTLD  scripts/kconfig/lxdialog/lxdialog
scripts/kconfig/mconf Config.in
#
# using defaults found in .config
#

*** End of configuration.
*** Execute 'make' to build the project or try 'make help'.

vadim@vadim-PC:~/Repos/busybox-1.36.1$ make -j$(nproc)
SPLIT   include/autoconf.h -> include/config/*
GEN      include/bbconfigopts.h
GEN      include/common_bufsiz.h
GEN      include/embedded_scripts.h

```

Однако процесс сборки завершился с ошибкой из-за наличия нескольких неопределённых символов:

```

networking/tc.c: In function 'cbq_print_opt':
networking/tc.c:236:27: error: 'TCA_CBQ_MAX' undeclared (first use in this function); did you mean 'TCA_CBS_MAX'?
236 |         struct rtattr *tb[TCA_CBQ_MAX+1];
    |                             ^~~~~~
    |                             TCA_CBS_MAX
networking/tc.c:236:27: note: each undeclared identifier is reported only once for each function it appears in
networking/tc.c:249:16: error: 'TCA_CBQ_RATE' undeclared (first use in this function); did you mean 'TCA_TBF_RATE64'?
249 |         if (tb[TCA_CBQ_RATE]) {
    |                ^~~~~~
    |                TCA_TBF_RATE64
CC      miscutils/ttysize.o
CC      miscutils/ubi_tools.o
networking/tc.c:255:16: error: 'TCA_CBQ_LSSOPT' undeclared (first use in this function)
255 |         if (tb[TCA_CBQ_LSSOPT]) {
    |                ^~~~~~
networking/tc.c:256:61: error: invalid application of 'sizeof' to incomplete type 'struct tc_cbq_lssopt'
256 |         if (RTA_PAYLOAD(tb[TCA_CBQ_LSSOPT]) < sizeof(*lss))
    |                                                         ^
networking/tc.c:261:16: error: 'TCA_CBQ_WRRROPT' undeclared (first use in this function)
261 |         if (tb[TCA_CBQ_WRRROPT]) {
    |                ^~~~~~

```

На ядрах линукса начиная с 6.8 в заголовочном файле `include/uapi/linux/pkt.sched.h` было удалено несколько символов, связанных с traffic control, из-за чего сборка `tc.c` проваливается. <https://lists.busybox.net/pipermail/busybox-cvs/2024-January/041752.html>

Поэтому из конфигурации Busybox уберём tc:

```
Networking Utilities
Arrow keys navigate the menu. <Enter> selects submenus --->.
Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes,
<M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </>
for Search. Legend: [*] built-in [ ] excluded <M> module < >
↑(-)
[*] ping (10 kb)
[*] ping6 (11 kb)
[*] Enable fancy ping output
[*] pscan (6 kb)
[*] route (8.7 kb)
[*] slattach (6.2 kb)
[*] ssl_client (25 kb)
[ ] tc (8.3 kb)
[*] tcpsvd (14 kb)
[*] udpsvd (13 kb)
[*] telnet (8.8 kb)
[*] Pass TERM type to remote host
[*] Pass USER type to remote host
[*] Enable window size autodetection
↓(+)
```

```
<Select>    < Exit >    < Help >
```

И повторно запустим сборку:

```
AR      libbb/lib.a
LINK    busybox_unstripped
Static linking against glibc, can't use --gc-sections
Trying libraries: m resolv rt
Library m is needed, can't exclude it (yet)
Library resolv is needed, can't exclude it (yet)
Library rt is not needed, excluding it
Library m is needed, can't exclude it (yet)
Library resolv is needed, can't exclude it (yet)
Final link with: m resolv
vadim@vadim-PC:~/Repos/busybox-1.36.1$ file busybox
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically
linked, BuildID[sha1]=f78ade8362f26698c4924f63cf137e679943ee45, for GNU/Linux 3
.2.0, stripped
vadim@vadim-PC:~/Repos/busybox-1.36.1$
```

Выполним команду `make install`:

```
./_install//usr/sbin/ubiupdatevol -> ../../bin/busybox
./_install//usr/sbin/udhcpd -> ../../bin/busybox

-----
You will probably need to make your busybox binary
setuid root to ensure all configured applets will
work properly.
-----

vadim@vadim-PC:~/Repos/busybox-1.36.1$
```


Собираем initramfs:

```
vadim@vadim-PC:~/Repos/busybox-1.36.1/_install$ ls -l
total 12
drwxrwxr-x 2 vadim vadim 4096 May 30 20:02 bin
lrwxrwxrwx 1 vadim vadim 11 May 30 20:02 linuxrc -> bin/busybox
drwxrwxr-x 2 vadim vadim 4096 May 30 20:02 sbin
drwxrwxr-x 4 vadim vadim 4096 May 30 20:02 usr
vadim@vadim-PC:~/Repos/busybox-1.36.1/_install$ find . | grep -v './initramfs.cpio.gz' | cpio -o -H newc | gzip > initramfs.cpio.gz
3109 blocks
vadim@vadim-PC:~/Repos/busybox-1.36.1/_install$ ls -ld initramfs.cpio.gz
-rw-rw-r-- 1 vadim vadim 1073190 May 30 20:14 initramfs.cpio.gz
vadim@vadim-PC:~/Repos/busybox-1.36.1/_install$
```

Запускаем прошивку:

```
vadim@vadim-PC:~/Emulation/arm-vexpress$ QEMU_AUDIO_DRV=none qemu-system-arm -M
vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -
append "console=ttyAMA0 rdinit=/bin/ash" -nographic
[ 0.000000] Booting Linux on physical CPU 0x0
[ 0.000000] Linux version 6.18.33 (vadim@vadim-PC) (arm-linux-gnueabi-hf-gcc (
Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #
1 SMP Fri May 29 15:29:39 +07 2026
[ 0.000000] CPU: ARMv7 Processor [410fc090] revision 0 (ARMv7), cr=10c5387d
[ 0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT nonaliasing instruc
[ 1.998630] Freeing unused kernel image (initmem) memory: 2048K
[ 2.002426] Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ # [ 2.099708] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/b
us@400000000/bus@400000000:motherboard-bus@400000000/bus@400000000:motherboard-bus@4
00000000:iopfpga@7,00000000/10007000.kmi/serio1/input/input1
[ 2.213478] input: AT Raw Set 2 keyboard as /devices/platform/bus@400000000/bu
s@400000000:motherboard-bus@400000000/bus@400000000:motherboard-bus@400000000:iopfpga
@7,00000000/10006000.kmi/serio0/input/input2
[ 11.990671] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 11.990958] amba 100e0000.memory-controller: deferred probe pending: (reason
unknown)
[ 11.991014] amba 100e1000.memory-controller: deferred probe pending: (reason
unknown)
[ 11.991048] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
ls
bin dev linuxrc root sbin usr
~ # echo 'Hello world!'
Hello world!
~ # exit
[ 79.835504] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00
000000
[ 79.844322] CPU: 0 UID: 0 PID: 1 Comm: ash Not tainted 6.18.33 #1 NONE
[ 79.844797] Hardware name: ARM-Versatile Express
[ 79.845086] Call trace:
[ 79.845862] unwind_backtrace from show_stack+0x10/0x14
[ 79.846816] show_stack from dump_stack_lvl+0x54/0x68
[ 79.846989] dump_stack_lvl from vpanic+0xc4/0x2f0
[ 79.847124] vpanic from __do_trace_suspend_resume+0x0/0x4c
[ 79.847844] ---[ end Kernel panic - not syncing: Attempted to kill init! exit
code=0x00000000 ]---
```